

CRITICAL THINKING ABOUT RESEARCH

PSYCHOLOGY AND RELATED FIELDS

SECOND EDITION



JULIAN MELTZOFF AND HARRIS COOPER

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Foreword

My father, Julian Meltzoff, loved research. He enjoyed mulling over research designs to think about what could—and could not—be inferred from any given study. He had a special gift in this regard, a “research eye” that could detect the slightest flaw in any research article. Some of us can look at a wall of pictures and immediately see one that is crooked amidst a dozen that are not. Other people can hear a wrong note played in a sea of music, or feel minute roughness on an otherwise smooth surface. My father could rapidly detect fatal flaws in research design and gaps in critical thinking that eluded peer review. His ability derived from years of training and a highly logical mind.

Julian Meltzoff believed that this ability to detect errors didn’t come naturally but could be cultivated and enhanced through education. When he sat down to write the first edition of *Critical Thinking About Research: Psychology and Related Fields*, he said he wanted to help others learn the value and joy of critical thinking. He also knew that critical thinking was becoming ever more important, as rapid communication and 24-hour news brought a tsunami of information to scientists and the public. What can one do to help people sort truth from fiction? His answer was to write a book to help others to learn to think critically about scientific and everyday information.

Research was exciting and fun for my father, and he wanted to make it fun for students. Readers of this book will see this come alive in the practice articles that constitute Part II. The careful reader will not only find the errors he instructively hid inside the articles but also will enjoy the amusing names he gave the authors of these papers. For example, the author of the article on childhood aggression is Junior Fites, an author on the article about the effects of pleasant experiences on time estimation is Bontempo, and an author of the article about self-talking is Earl Singleton. There are other buried treasures in the reference lists that will make readers smile as they labor to learn about critical thinking. My father’s goal was to show that learning about research could be fun, and that every word counted. Let nothing go unexamined.

Julian Meltzoff was a fellow of the American Psychological Association (APA) for 58 years. He published his first peer-reviewed paper on clinical

psychology in an APA journal in 1952 at the age of 31. He published his last peer-reviewed paper in an APA journal in 2014, at the age of 93. This last article, published in *Psychology of Aesthetics, Creativity, and the Arts*, is a unique piece that unites art and science. It focuses on an intriguing error that has been made by artists throughout history in depicting human gait. Sculptors and painters, including masters such as Rodin and Leonardo, often erred in how they placed the arms in relation to the legs when a person moved. Does the left arm swing forward or backward when the right leg moves forward in natural gait? My father showed that there is one correct answer, and that artists from ancient Rome through modern times did not always get this right. It is a breathtaking piece for a 93-year-old.

I am very pleased that APA has chosen to reissue my father's book and that Dr. Harris Cooper has updated it in ways that make it even more helpful to the modern student. Critical thinking in science and everyday life has, if anything, become more critical in this age of social media. We all need to think critically about not only scientific information but also everyday information. My father had long hoped that this book would make a mark by helping students studying behavioral sciences and everyday citizens learn the essential skills of critical thinking. Happily, with Dr. Cooper's contributions and additions, this classic APA book will now reach an even wider audience.

Andrew N. Meltzoff, PhD
Professor of Psychology
University of Washington

Preface to the Second Edition

It is an honor to have been asked to prepare the second edition of *Critical Thinking About Research: Psychology and Related Fields*. The author of the first edition, Julian Meltzoff, passed away at age 94 after years as a clinical psychologist, researcher, and writer. In the latter role, he wrote about psychotherapy and posttraumatic stress disorder, and he authored this book on how to read and critically evaluate psychological research. He also enjoyed a second career as a sculptor whose bronze piece “Body in Motion” graced the cover of the December 2014 issue of *American Psychologist*. Although I never met Dr. Meltzoff, the opportunity to refresh this book for the next generation of psychologists gave me an introduction to his scholarship, insight, and humor. I am most grateful for the opportunity.

I have tried, in the second edition, to remain true to Dr. Meltzoff’s voice, though the content now reflects both of our sensibilities. Part I of the book has been extensively revised to reflect the enhancements in methodology and methodological thinking that have appeared since the first edition. I also added a chapter on research synthesis and meta-analysis, techniques that were just emerging when Dr. Meltzoff wrote the first edition. I have, however, largely remained true to Dr. Meltzoff’s organizational structure. I revised the Glossary by adding terms, which are now italicized in the text when they first appear or play a prominent role in what is being discussed. That way, you can turn to the glossary if you need reminding of a term’s meaning and know the term will be there. I also have added a summary of the main points at the beginning of each chapter. Other than updating the (fictional) dates and some of the language in the practice articles in Part II, these remain largely unchanged. Dr. Meltzoff did an inspirational job with these, and even his choices of topics remain remarkably current. I added one new example of an article reporting a meta-analysis.

I also thank Linda Malnasi McCarter, acquisitions editor at the American Psychological Association (APA), for thinking of me when this opportunity became available. Beth Hatch, my development editor at the APA, orchestrated two excellent reviews and provided her own cogent

comments. My wife, Beth, put up with me brilliantly and sensitively, knowing when to leave me be and when to drag me away from my writing.

One more personal note. As I worked through the revisions, I realized I was spending hours “listening to” Dr. Meltzoff’s reaction to my changes. This led me to read some of his other writings, to ensure I was hearing him clearly. Most notable was his self-published book, *Memories of WWII*. I realized he and my father, Samuel Meyer Cooper, a high school graduate, served together in the European Theater of Operation during the war. They never met either.

Harris Cooper

Preface to the First Edition

This book was stimulated by my having spent the last 17 years teaching graduate students how to design and critique research. During this period, I had the responsibility of critically evaluating far more dissertation proposals than I care to admit, for fear of being too closely associated with Baron Munchausen.

Although there are many good texts on research design, they are written from the perspective of the research producer, not the research consumer. My colleagues and I have had to become inventive and innovative when teaching the important skill of critiquing research. From a pedagogical point of view, the need for supplying flawed research articles on which students could practice their critiquing skills was always apparent to me. The idea of creating the materials to order came later. Construction of such reading matter was informed by my experience in discovering errors that I repeatedly encountered when reviewing published articles, as well as research proposals. I am indebted to all of the students and professional psychologists who unknowingly supplied me with the ideas that went into this book, and who so graciously recognized and corrected the problems once they were revealed to them. My thinking about designing and critiquing research was enriched by frequent detailed exchanges of ideas with my academic colleagues. Their cogent observations, and sometimes their alternative ways of looking at things, stimulated me continuously to reexamine research-related issues, to fine-tune my views, and to reappraise my judgments.

I am most grateful to all of those who reviewed parts of this manuscript and offered helpful suggestions to improve its scope, its focus, and its clarity. I am particularly indebted to Mark Sherman, with whom I discussed research issues so frequently over the years. He helped shape my ideas about the subject and the ways it can be taught. He not only read the entire manuscript, but gave me detailed and expert feedback about it. I am truly grateful to Andrew Meltzoff for his frank, careful, and incisive critique and his ongoing interest and support in this effort. Without the constant encouragement and support of Antonia Meltzoff, I probably would not have been so inclined to

begin this project or to have become so engrossed in carrying it through to completion. As each section was converted from thought processor to word processor, it turned into a prolonged show-and-tell in which she participated with a gusto that fueled my own esprit. Nancy Meltzoff was a most helpful resource when I dared to venture into her area of specialized knowledge. If, with so many who share my surname involved, writing this book is beginning to sound like a cottage industry, it is only because all of them are PhDs who are highly accomplished scholars, scientists, writers, and educators in their own right, whose judgment I trust, whose candor I can depend on, and whose input I value.

I have Peggy Schlegel, acquisitions editor at the American Psychological Association (APA), to thank for the guidance and encouragement she provided from the beginning. My special gratitude goes to Andrea Phillippi, development editor at the APA, for her careful reading and most helpful suggestions. I am deeply grateful to Ida Audeh, whose formidable editorial skills have done so much to polish the text. I very much appreciate the work done by all of the reviewers and editors of the manuscript in its incipient and later stages.

Julian Meltzoff

CRITICAL THINKING ABOUT RESEARCH

Introduction

How should scientific evidence be evaluated? After reading a research article, one person may be fully convinced by the evidence that is offered; a second person may think that the same evidence might be true but regards it, on balance, to be equivocal and feels obliged to suspend judgment; and a third person may disdainfully dismiss it with scatological words of unmistakable meaning. Their different judgments may be informed by different views about what proof is. They may be influenced by their different beliefs about what evidence is needed to establish proof. They may be limited by the amount of knowledge that they have about the methods used to gather evidence in the search for truth. Finally, they may be affected by variation in their ability to think critically.

Regrettably, there is evidence that people reading the same research paper but beginning with opposing beliefs come away with even more polarized opinions than before they were exposed to the new evidence (Lord, Ross, & Lepper, 1979). This can happen because people misinterpret the evidence they have read, they are motivated to think in a certain way, or because of other cognitive malfunctions (McFadden & Lusk, 2015). This is a quality of human nature that is antithetical to the goals of science. It is what we attempt to overcome when we put on our critical thinking caps.

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Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper

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We use the phrase *ability to think critically* as a positive quality, not as a pejorative reference to the characteristics of a mean-spirited faultfinder. The phrase refers to the skill of thinking about an issue, analyzing it, looking at it from all sides, and weighing whether there is sufficient evidence of good-enough quality to warrant making a reasoned judgment that is as free of personal bias as possible. The Critical Thinking Community (2016) defined it this way:

Critical thinking is self-directed, self-disciplined, self-monitored, and self-corrective thinking. It presupposes assent to rigorous standards of excellence and mindful command of their use. It entails effective communication and problem-solving abilities, as well as a commitment to overcome our native egocentrism and sociocentrism. (p. 1)

Texts and curricula are available to teach critical thinking in general (e.g., Moore & Parker, 2015). Our purpose here is a bit more restricted. We focus on critical thinking in the context of evaluating scientific research. We hope (a) to exhort you to adopt a critical mental set when reading research literature; (b) to try to increase your understanding of the principles and methods of research that are used to produce evidence; (c) to alert you to booby traps that can compromise the research and cloud the evidence; and (d) to provide you with materials that can be used for practice, with the goal of improving and honing your critical reading skills when you are exposed to research in the sciences, especially psychology and related fields (education, law, economics, health, nursing).

This two-part book is about how to evaluate scientific evidence. Part I provides an overview of the methods and principles that you need to know about when you critique or evaluate the trustworthiness of research. Part II presents fictional journal articles that provide you with practice in applying the principles outlined in Part I. The book is certainly not a statistics text, and it assumes that you have some familiarity with basic statistical methods. It is not intended to be your only text to prepare you to design and to carry out research. The focus is on reading and critiquing research. The two activities, producing and consuming research, are in no way contradictory, but a person who is adept at critiquing research is not necessarily skilled at thinking up research questions or conducting research any more than a drama critic is necessarily a good playwright. Books on research methods aim to give people who are preparing to do research an insider's view of how to go about designing a study and planning and conducting statistical analyses of data.

This is not a stand-alone methods book for aspiring and practicing researchers, although they can certainly use it to hone their skills as they think about how to proceed. It is intended to assist you in reading and assessing research. A reader who may not be familiar with the intricacies of *research design* (and who may not be aware of all the pitfalls to avoid) should know enough to recognize procedures that have gone awry or to know when plans have not worked out optimally. We review the application of research principles and alert you to things that interfere with valid research. Our objective is to help you bring an inquiring and informed mind to the evaluation of research.

Our presentation of the ways an experiment can flounder, and the emphasis we place on scrutinizing and critiquing research, is in no way intended as an indictment of the research process or as an expression of any reservations whatsoever about its value as a method of seeking the truth. In fact, we know that the perfect study has never been done. All research has strengths and flaws. Your task as a critical reader is to identify these and judge the value of the research for making an informed judgment of its utility in drawing the conclusions the authors intended.

We hope to encourage a form of interactive reading that we believe is central to the process of critical evaluation of research literature, as opposed to the almost reverential acceptance (or rejection) of the printed word. You must be able to bring a skeptical, inquiring, searching, probing attitude to the task; to spot things that do not look quite right; and to recognize how a weakness in one phase of the design or execution of a study can severely compromise the study at a later phase. The “tests” take the form of questions that you, the critical reader, ask of the written research report.

Why Psychologists and Professionals in Related Fields Need to be Critical Thinkers

Psychologists may spend their time in one of three ways: (a) doing research to generate knowledge, (b) transmitting knowledge to others by teaching or by directing the research of others, or (c) applying psychological knowledge in the form of clinical or consulting services. These can be done singly or in any combination. A parallel set of options is available to specialists in other behavioral or social sciences, in education, in law, in medicine and the health sciences, and in the service of organizational improvement. There is variation between and within fields in how much time is spent educating and training students for the functions of generating knowledge, disseminating it, or applying it.

One thing shared by people in psychology and all related fields and subspecialties, no matter what career direction they choose, is that they all must be consumers of research. The bedrock of psychology and related fields consists of theories and data from research that support or refute the theories. All who profess to know a field, and those who want to keep abreast of advances, are necessarily consumers of research. It stands to reason that they must be able to evaluate research critically. Having a good grasp of the scientific method and the principles of research design, and knowing what to look for, are invaluable in this pursuit.

Psychologists who graduate from research-oriented doctoral programs have the advantage in this regard because research knowledge is a vital part of their education and the dissertation requirement gives them hands-on experience in research. Nevertheless, virtually all programs that do not require a research dissertation strive to teach students to think critically and to read empirical work intelligently. Nobody who is considered an expert or specialist likes to be the bearer of misinformation. Throughout the history of psychology and related fields, untested fads, half-baked ideas, and outright quackery have led astray theories, policies, and practices. Many

professionals with weak skills for critically evaluating research have become enthusiastic advocates of unsubstantiated claims. Other professionals have had the wherewithal to resist and have retained their skeptical reserve until solid research data were furnished.

What is necessary, then, is first to learn how to critique research and then to make a habit of sifting all incoming offerings through a sieve whose openings are small enough to permit only the trustworthy evidence to fall through. This two-part book is written to further that goal. We present the principles of research design that you need to know about to critique research (Part I), followed by intentionally flawed, but realistic, fictitious materials on which to practice your critiquing skills (Part II).

Overview of the Chapters

Part I consists of 11 chapters. Chapter 1 deals with the methods of seeking knowledge about our world, the rules of scientific evidence, and what constitutes scientific proof. It discusses how to read critically and presents an outline to serve as a guide for critiquing research. Chapter 2 focuses on the questions and *hypotheses*, or predictions, that motivate research and discusses faulty reasoning about causation. Chapter 3 reviews the critical appraisal of research strategies and the selection of variables. The selection and composition of the research participants, or *sample*, methods of sampling, and the consequences of faulty sampling are covered in Chapter 4. Chapter 5 gets to the heart of research design in a discussion of the variables that should be under the control of the researcher and methods of controlling them. Chapter 6 presents the major *experimental* and *quasi-experimental designs* that are available, and discusses the flaws that you should recognize in some of them. It also gives examples of some of the important threats to the trustworthiness of a study. Chapter 7 discusses the selection of *independent variables*¹ and *dependent variables* (or outcomes) to help you appraise whether the selected techniques were appropriate and used properly in a particular research study. Chapter 8 provides an orientation to the assessment of statistical analyses, graphic analysis, and the drawing of inferences and conclusions. Chapter 9 looks at how data are interpreted and statistics translated to information. It also covers issues to consider when evaluating the completeness and transparency of a report of research. Chapter 10 introduces a different kind of research, the *research synthesis*, or *meta-analysis*, in which the goal of the researchers is not to collect new data but to accumulate and interpret the studies on a topic that have already been conducted. Ethical considerations in research, the subject of Chapter 11, are discussed to alert you to deviations from ethical standards. All the chapters in Part I are written from the perspective of readers looking in on the research report from the outside, with

¹Most often, we use the term *independent variable* to designate a variable that is being manipulated in an experiment, and therefore is viewed as a cause. The term *predictor* is used in a nonexperimental study, when the variable is measured. *Outcome* or *criterion variable* is used in a study of a treatment or intervention. We will try to use these terms in this manner through the text.

the emphasis on critically evaluating research rather than conducting research on their own.

Part II consists of 17 realistic but short and entirely fictitious simulations of journal articles. The articles have flaws built into them that range from the blatant to the subtle. The collection is essentially a workbook that is based on the principles in Part I and is intended to be used by you to practice your critiquing skills. The frontispiece by the great 18th-century artist, Hogarth, was one of the things that instigated the idea to write this book. The legend at the bottom of the print says, “Whoever makes a design without the knowledge of perspective will be liable to such absurdities as are shown in this frontispiece.” So it is with research. Lack of understanding of the principles of research design can lead a researcher to the kinds of errors that are illustrated in the fictitious articles and can prevent you from realizing that there is anything wrong.

The articles in Part II range across a variety of content areas (e.g., clinical psychology, general psychology, psychology and the law, education, sociology) and research methods, but the predominant emphasis is on psychological experiments. Each fictitious article is accompanied by a critique of its built-in flaws. Critical issues are cross-referenced to the chapter in Part I that addresses them, so that if you want a fuller explanation you will know where to find it.

The kinds of statistical analyses in the articles range from simple descriptive statistics to univariate *analyses of variance* with *categorical variables*. Studies that use the techniques listed in the fictitious articles are abundant, and the articles mirror those that can be found in the literature. The articles do not feature highly advanced statistical techniques, so that the book will be useful for the greatest number of people.

These critiquing exercises are meant primarily for the use of students in psychology and related fields, whether they be advanced undergraduates or beginning graduate students. Psychologists whose graduate school experience is well behind them, and who have not themselves been engaged in research, can profit from these as well.

Because the principles of research design and the approaches to critiquing research are not restricted to psychologists or psychological research, students and professionals in other behavioral and social sciences also should find these materials and exercises useful. This book may be helpful to medical researchers, consumers of medical research, educators and education administrators, and members of the business and legal profession who deal increasingly with the evaluation of evidence and opinions that are offered by scientific expert witnesses. With so many scientific specialties, cross talk among them is necessary to function intelligently. Your understanding of research in another field may be circumscribed by a lack of technical knowledge, but you should still be able to read and understand research in a related field.

In this age of science, the media continuously expose nonscientists to science reports from all fields. The ordinary person is served a rich diet of science every day. The portions, unfortunately, take the form of sound bites that must be swallowed whole or of summaries that lack all the details. Efforts to facilitate public engagement with science are plentiful and growing more so (e.g., Alan Alda Center for Communicating Science, 2016; Center for Engagement With Science and Technology, 2016). In psychology, the call for practices to be evidence based is taking hold throughout

the profession (e.g., American Psychological Association Presidential Task Force on Evidence-Based Practice, 2006). Educated nonscientists who want to learn more about research design and how to read research critically could profit from reading in Part I, perhaps skipping sections that appear to be too technical and specialized. As information consumers, they could derive benefit from testing their critical reading skills on the exercises in Part II.

Who Will Find This Book Useful?

Readers who are advanced undergraduate or graduate students may read this book as part of a course's materials, or can use it to assess their own level and needs. Those who have already had considerable training in research design should select any sections of Part I that will fill the gaps. You should evaluate the articles in Part II on your own before reading the accompanying critiques. Concepts that are relevant to the article are cross-referenced with Part I; you should consult Part I for further clarification.

Professional psychologists who have not done any research or who have done some research in the past but have been away from it for a long while may choose to look through the contents of Part I and start to read wherever they see a need for a refresher or wherever something strikes their interest. Those who evaluate the articles in Part II independently and find that they have missed many of the central points in the critiques may benefit from reading Part I more thoroughly.

Readers who believe that they already have a good understanding of research questions, hypotheses, and sampling but that they could use a booster on design options and methods of control might want to read Chapters 5 and 6. Those who are comfortable with various aspects of design but have questions about independent variables and dependent measures could start with Chapter 7. In either case, you should try to sharpen your reading skills by critiquing the articles in Part II.

Knowledgeable and experienced researchers who already know far more than is covered in Part I may begin with Part II and read the fictitious articles for fun, for challenge, and for mental calisthenics.

PRINCIPLES OF CRITICAL THINKING



Seeking Trustworthy Knowledge

1

A critical reader of research

- is skeptical of scientific claims on the basis of (a) beliefs that have simply been held a long time, (b) appeals to authority, (c) votes, (d) rational arguments alone (especially if the premises have not been tested), and (e) observation and sensory input (because of the possibility of perceptual distortions);
- understands that because evidence is written down does not mean it is correct;
- is aware that publishing in science has corrective devices (e.g., peer review) that protect against errors and false claims, but this system is not perfect—the Internet, in contrast, has weak, if any, mechanisms for preventing error-filled and fake research reports from being published;
- uses an interactive mode of thinking, which involves active and frequent use of explicit rules to judge whether expectations are met;
- has clear and justifiable reasons when an evaluative judgment is made about the strengths and limitations of an article;
- knows the ability to learn from science requires knowledge of research design and the skillful application of that knowledge; and
- maintains an attitude that the researchers are responsible for providing a complete and accurate rendition of their work.

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Critical Thinking About Research: Psychology and Related Fields, Second Edition, by J. Meltzoff and H. Cooper

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Critiquing research is not something that comes without learning and practice. We spend many years applying other ways of appraising truths before we learn the scientific method. These modes of thought continue to influence us in deciding what evidence to consider valid, even after we have learned the scientific method.

In this chapter, we review (a) the various methods people use for seeking trustworthy knowledge about their world, and (b) differing views on what constitutes sufficient evidence for something to be taken as “proven.” If you aspire to be an informed citizen who makes sound decisions, you need to look on truth claims with a critical eye. The crux of the critical appraisal, in turn, depends on the rules of evidence that you hold a truth claim up against. When scientific evidence is being evaluated, the appraisal rests on your knowledge of research design and how carefully you apply that knowledge.

Methods of Seeking Knowledge

People use many different methods to seek knowledge. These methods may be based on rules of evidence that conflict, coexist, or complement each other. The same individual may use different rules of evidence in different spheres of life. Some people can switch effortlessly from one mode of thought to another (e.g., when you read a historical novel, interact socially, go to church, read scientific studies). Other people find switching the criteria for truth more difficult. They may apply the wrong rules to the circumstance in which they find themselves. A trusted friend may be a good source of information about the quality of food in a restaurant but may be less reliable when it comes to imparting historical facts.

Think about a simple statement you could make that you are convinced is true. Got one? Okay, now think about how you know it to be true. Charles Sanders Pierce (1877), a 19th-century American philosopher and scientist, suggested that we know things based on several sources of evidence. We suspect your truth will be based on one of these.

HABIT OR TENACITY

First, Pierce (1877) suggested that we know things because they appeal to us, we repeat them to ourselves often, and we turn away from any information that might disturb our happy state of affairs. Although we might be denying contradictory facts, we find the state or certainty that tenacious beliefs bring us more satisfying than the unsettled state of mind that accompanies doubt. If we do this long enough, we might forget where and when the original belief took hold, but no matter, “I’ve believed it to be true for so long, it must be so.”

Clearly, this method of knowing has its failings. We will meet people with different facts and discover that we have no way of determining whose facts are more accurate. Likewise, over time, the world will change and our beliefs may become dysfunc-

tional, leading us to act in ways that do not have the consequences we anticipated. For example, if your truth is, “The climate of the earth is not growing warmer,” you may have forgotten when and how you learned this, but the consequences may be dire of holding on to the belief simply because you have always believed it.

AUTHORITY

Another way we know things to be true is to take the judgment of one or more authorities as sufficient evidence. Pierce (1877) emphasized the point that “truth by authority” often has a coercive component to it. He felt that religious and political institutions can promote a way of thinking about the world. Our parents, friends, and teachers are also sources of evidence whose authority we accept. We also can view experts as authorities, whose trustworthiness we believe because their credentials make them credible sources of information. Experts can be highly reputable scientists whose education and experience in designing and conducting experiments give credibility to their pronouncements. So, if your truth is, “blessed are the meek for they shall inherit the earth,” you probably learned this from the Bible or from a religious authority. So too the truth “hydrogen and oxygen combine to make water” was probably learned from an authority, your first chemistry teacher.

But what if two or more experts disagree, or experts change their mind? With the unerring benefit of hindsight, one can see how mistaken highly qualified scientific authorities have been in their pronouncements. In *The Experts Speak: The Definitive Compendium of Authoritative Misinformation*, Cerf and Navasky (1998) provided retrospectively amusing quotes from outstanding authorities in various fields. The physicist Lord Kelvin, who was president of the British Royal Society, announced in 1895 that heavier-than-air flying machines were impossible. Albert Einstein is reported as having said in 1932 that harnessing the energy of the atom would never be feasible. In 1956, the leading British astronomer Richard van der Riet Wooley alleged that space travel was nonsense. Clearly, there were scientists who disagreed, and if we could ask them today, Kelvin, Einstein, and van der Riet Wooley would regret what they said.

DEMOCRATIC JUDGMENT

Trust in the judgment of many, in contrast to the dicta of single authorities, also is a common way of deciding what is right. Pierce (1877) wrote, “Let the actions of natural preferences be unimpeded, then, and under their influence let men, conversing together and regarding the matter in different lights, gradually develop beliefs in harmony with natural causes” (p. 8). We become confident about the truth-value of statements when a judgment is shared by many people. But consensus is not science. While Pierce had a clear affection for this method of knowing, he also recognized that even democratically arrived-at truths had their failings. They are subject to fashion and thus, rather than being eternal truths, could be prone to swings based on the tenor of the times. Groups can also engage in “groupthink,” the process whereby

group pressure leads to the suppression of the views of minorities and the reality testing of ideas (Janis, 1972).

REASONING

Another way of arriving at truth is by reason alone. Under certain circumstances pure reason can lead to what is called *necessary truth*. In *a priori reasoning*, conclusions are deduced from known or self-evident principles or definitions. If your truth is “ $1 + 1 = 2$,” you have invoked a self-evident truth. The *syllogisms* of formal logic and the proofs of pure mathematics lead to necessary truth.

However, not all questions can be answered by pure reason alone. Many conclusions about the world call for *inductive reasoning* (i.e., reasoning that begins with things taken as known). Final judgments are based on experience, and principles are generalized from facts that are established by experience rather than by the application of reason alone. If the observations are wrong (the premises of a logical argument), then so are the conclusions.

SENSORY INFORMATION

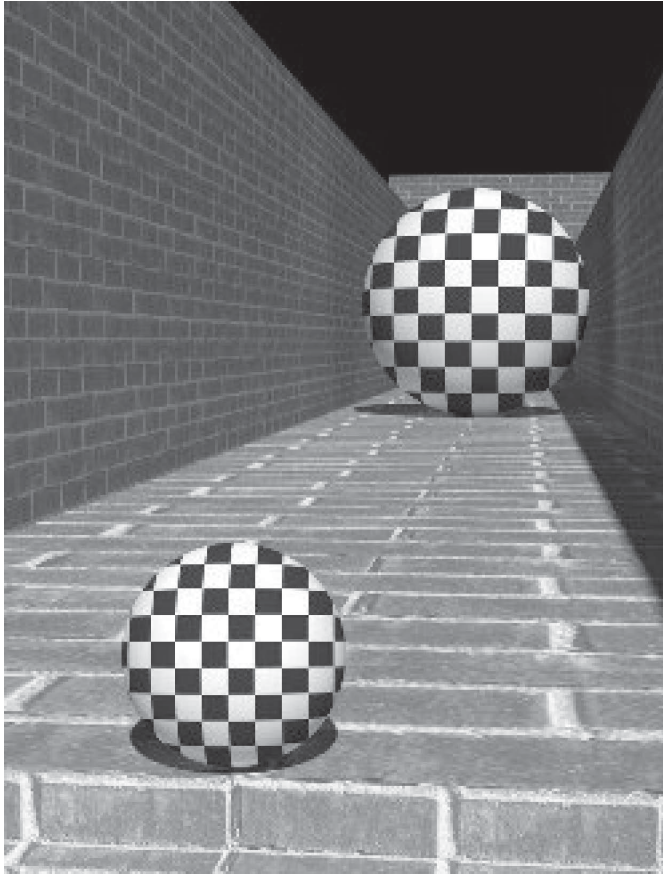
This reliance on experience for finding knowledge leads to another method of knowing: One of the most compelling and unshakable proofs for people is personal experience. There is a certainty to the subjective truth of the statement “I have seen it, I have experienced it. Therefore, I know that it is true.” If you answered our request for a simple statement you know to be true with “I have two eyes,” it is because you have looked in the mirror day after day and have seen your two eyes staring back at you.

Yet, we know the human senses are fallible. Psychologists have studied optical illusions, and magicians have exploited this fallibility to their advantage. For example, look at the soccer balls in Figure 1.1. Although the ball at the top of the figure clearly appears larger, the two balls are the same size. (Not only does the optical illusion show us why our senses cannot always be counted on to reveal truth, but it can also be used to question the veracity of democratic judgments; if we voted, we would unanimously agree that the ball at the top of the figure looks larger.)

EMPIRICAL AND EXPERIMENTAL METHODS

Far and away, according to Pierce (1877), the best way to know things is through the methods of science:

It is necessary that a method should be found by which our beliefs may be determined by nothing human, but by some external permanency—by something upon which our thinking has no effect . . . the method must be such that the ultimate conclusion of every man shall be the same. Such is the method of science. (pp. 9–10)

FIGURE 1.1

Illusion of soccer ball size. From <http://www.123opticalillusions.com/pages/sphere-size-illusion.php>

We can think of the scientific method as one that combines rationality and observation. At its core, scientific knowledge is based on the application of objective and systematic rules to observations or experiments. In experiments, the scientists exercise control over the circumstances under study by careful prearrangement of conditions. In nonexperimental empirical work, the scientists observe, measure, and record but do not control the introduction of the *variables* (when the scientist does control the introduction, these are called *manipulated variables* or *independent variables*).

Let us take a scientific approach to testing the assertion that the top soccer ball in Figure 1.1 is larger than the bottom one. We do not take our observation for granted but instead test it systematically by finding the center of each ball and measuring its circumference with a protractor, a measurement instrument that is objective, unbiased, and not reliant on any person's subjective experience. The measurement

reveals that the balls are of equal size. As good scientists, we then write up our experiment, being careful to fully and accurately describe what we did so that others can repeat our study if they wish, so that the conclusion of every observer will be the same.

The scientific approach to discovering or establishing knowledge reflects a basic skepticism about proof and is characterized by a “show me,” “prove it,” and “what is the evidence?” attitude about alleged facts. Your truth is just yours until you demonstrate it through means external to yourself and open for scrutiny (and critique) by everyone.

The scientific method also accepts a degree of uncertainty to all its conclusions. Mathematics are used to measure things and statistical tests are applied to results so we can get a sense of how dependable or unreliable the conclusions are. Although any conclusion that scientists reach is never an objective certainty, you can judge how probable it is, through statistical analysis and *replication*, the repeating of the experiment. Even though absolute certainty may be beyond reach, scientists do aspire to attach a level of probabilistic confidence to judgments about what is true. Science encourages an open mind as well as a skeptical mind; implicit in the claim of scientists is the proviso “I may be wrong, but . . .”

The scientific method is the best way to produce knowledge that is unbiased, exists outside any given individual, is open to and testable by others, and has the best chance for permanency and alteration (if the evidence or conditions change). However, it is not the principal way we know things. In fact, if we could count the ways you answered our request for a truth that appeared at the start of this chapter, we would discover that the methods of observation and authority would be the methods used most frequently. Although you may have stated a scientific fact as your truth, in a vast number of instances this fact would be known to you because you read it in a book (or heard it from a science teacher), not because you ran the science experiment yourself. Therefore, the source of your knowledge was an authority, in this case the authority of a science textbook whose contents you trusted. Over time, the written word has grown more and more important in the transmission of knowledge (and falsehoods) and has been joined by video as the way we are most likely to learn about our world. The written word is the principle mechanism for communication of science.

REVERENCE FOR THE WRITTEN WORD

With the advent of the Internet, communicating through writing experienced a revolution unparalleled since the invention of the printing press. The Internet has made it possible for writers to communicate with audiences of enormous size in a matter of seconds. And the cost of communication, the need for writing equipment, and the need for a place to store documents have all diminished greatly.

A few things stand out. First, the ability to write and circulate what is written has been democratized. Anyone with access to a computer and the Internet can do it. But although it is certainly simpler to record ideas and to communicate them to others, this does not mean worthwhile thoughts have become easier to recognize. If anything, the

ability to discern what to believe and what to disregard has grown more difficult and more important. Even videos are open to corruption, through image manipulation and outright lying. On the Internet, you cannot always believe what you read and see.

When almost anyone can disseminate words and thoughts, it becomes critical that you have the skills to separate the wheat from the chaff. This sets the stage for critical reading and makes it a more essential thing for you to do. And because scientific discovery requires an initial skepticism, it is the perfect place to hone your skills as a critical reader.

What Critical Reading Requires

Any critique of what is presented as a scientific study addresses the way the study does or does not meet the scientific standards for evidence and proof. In evaluating such a study, you, as a critical reader, focus on its scientific soundness, not on whether the findings conflict with your preexisting faith, beliefs, or ideas about social acceptability; not on whether the results conflict with expert opinion or clash with other methods of establishing truth and gathering evidence; and not on whether the results are in harmony with your gut feelings. Our soccer ball example in Figure 1.1 shows that one scientist with a protractor can overturn the evidence from our senses or from an otherwise unanimous group.

Even though it cannot provide more than probable truth and must be taken with caution, *empirical* (and particularly experimental) research is still the best way to look for answers to certain types of questions. The emphasis in this book is on the ways an experiment can fail. Scrutinizing and critiquing research is in no way intended as an indictment of the scientific process or as an expression of reservation about its value as a method of seeking the truth. In fact, it is the skepticism of scientists toward their own work that makes their work so valuable.

Doing research and reading research reports critically are arduous processes. It is much easier to say, “This is too hard, let’s just brainstorm it and take a vote,” or “Let’s just ask Dr. Jones, he ought to know.” People may become so overwhelmed with all of the difficulties associated with doing research that they detach themselves from it and declare that they are simply not interested in using this method of finding things out. But becoming a skilled reader of research is no different from becoming skilled at anything; it takes time, effort, and practice. Some people dislike the skepticism of science and its assertion that all claims of knowledge are probabilistic; scientists hold assertions with different levels of certainty and might find tomorrow they are wrong. But other people (including you, we hope) would rather hold a true belief with some uncertainty than hold a wrong belief without doubt. Others who have no intention of actually conducting research still feel it is too difficult and too technical for them to even judge by themselves the value of other people’s work. They prefer to leave it up to the researchers, who, after all, must know best. They then take, on faith, whatever is printed in the scientific journal. This defeats the whole purpose of presenting research for public scrutiny. It is this attitude we hope to defeat with this book.

The phrase “research shows” is as revered in some circles as pronouncements of authorities are in others. The important thing for the person uttering these words is to have read the research, to have understood it, and to have evaluated it before using its findings to support a position or to inform a practice. When the research is truly sound, however, there is no more powerful way to make a point than to cite research findings.

Critical reading requires a mental set of a particular kind. This mental set can be taught, encouraged, and nurtured. Conversely, it can be discouraged or even forbidden. What is involved is a kind of general open skepticism that enables you to bring a “show me the evidence” attitude toward reading regardless of how authoritative the author may be or how attractively the words are packaged. You, as the critical reader, then engage in an interactive dialogue with the written words. You focus on each assertion that is written and contemplate the meaning behind it. You also think about what was *not* written by the researchers and wonder why it is absent. This cannot be done in haste, and it often requires multiple rereadings and rumination about particularly important or troublesome aspects of the research report.

Critical reading of scientific materials is very different from recreational reading of science fiction or of a novel (although these too can be read critically by literary scholars). When reading scientific documents, every thought that is expressed, every conclusion that is reached, must be screened through your internal skeptical scanner. If everything meets the most stringent cognitive challenges, you come away enriched and gratified. On the other hand, you may accept some of it but have reservations about other parts. A third scenario is that you may reject it in its entirety and resent having spent the time on it. (You can feel gratified, as well, by finding hidden weaknesses in a study, by the way.) None of these scenarios is the same as reading in a passive-receptive mode, with your critical antennae down, and accepting the author’s conclusions wholesale.

Uncritical acceptance of conclusions leads to the incorporation of misinformation into your body of knowledge. It is no trivial matter when such material misinforms public policy or becomes a false guide to professional practice. “Research shows” is one of the favorite expressions of experts (a) who are called on by the media to express their professional opinions on a wide range of topics, (b) who are asked to consult with or testify before lawmakers about social issues that affect public welfare, or (c) who are relied on to give counsel to other service providers or to educators.

Research psychologists carry a heavy burden of responsibility for ensuring the accuracy of their claims about their results. In turn, psychologists who cite or apply the research findings of others share that responsibility. They have an obligation to use their critical reading and evaluation skills in reviewing a study before they cite it as evidence that supports a point of view and before they apply the findings in their professional work. Uncritical acceptance of invalid research can impede the development of the field and jeopardize human welfare.

The process of critical reading is far different from that of gaining scientific information from summaries or digests of research results, as busy professionals are often tempted to do. These summaries or digests provide conclusions to be accepted on the reader’s faith in the expertise and impartiality of the person who prepared the

summary. But nobody has the time to read all the scientific and professional journals in their field. Digests and abstracts do serve a useful purpose. The best way for you to use them is as a screening device for subject matter that may be of special interest, to be followed by critical reading of the actual articles. There is an exception here. In the relatively recent past there has been an explosion in the amount of scientific research. To accommodate this, researchers have developed a set of rules for accumulating evidence on the same hypothesis (e.g., “What does all the research say on the relative effectiveness of different methods for teaching reading?”) in a complete, transparent, and unbiased manner. This approach is referred to as *systematic review*, *research synthesis*, or *meta-analysis*. We introduce you to these techniques in Chapter 10.

When faced with an original full-length article, experienced critical readers often skim through it quickly so that they can get a general overview of the contents and can decide whether they are sufficiently interested to read it more carefully. If they decide to engage, they then begin to reread the article in a fully-focused and critical mode. Sometimes several readings of crucial sections are required. This is particularly true when the text is fuzzy or lacks detail.

CRITICAL READING

The passive reader is a recipient who takes in what the text says. At the end, the passive reader may have a general impression of the research and some points of agreement or contention. On the other hand, the critical reader is interactive, actively anticipating what is to come. In scientific reading, this anticipation involves the use of rules for conducting valid research. These rules are used like yardsticks that are applied when they are relevant to the narrative about the study. The critical reader then discovers whether these expectancies have been met (“Does the research ‘measure up’ to this standard?”). Adoption of the critical, interactive mode places you in the best position to evaluate the work critically. Throughout this book, we call attention to the kinds of expectancies that flow naturally from given antecedents. These include the following:

- the research question guides the review of previous literature that sets the context for the scientific report;
- the literature review and the statement of the problem relate clearly to the hypotheses;
- the hypotheses set up research design expectancies and suggest what variables should be manipulated, measured, and/or controlled (so as not to bias the research outcomes);
- the hypotheses, appropriate design, and type of data dictate the method of data analysis; and
- the analysis of the data influences the kinds of conclusions, inferences, and generalizations that can be made.

The critical reader is constantly engaged, checking his or her evolving expectancies against the actual way the study was carried out. Through such interactive reading, the critique, whether negative, praising, or a bit of both, can be focused and justified.

Readers select articles to read because the title or *abstract* suggests it might be of relevance to the answers they seek. Think of a journal article with the title “Effect of Using Phonics and Whole Language Teaching Methods on Reading Ability.” You expect to find a cause–effect experiment in which the method of teaching is systematically manipulated and reading ability is compared in groups of children taught by the two different methods. The review of the literature and the statement of the problem reinforce that expectation, and the hypotheses are worded using causal language. Turning to the Method section, you are soon disappointed to find that instead of manipulating teaching methods systematically, the researchers located different elementary school classrooms within a large school system. The method of teaching reading varies from classroom to classroom. You immediately wonder about how similar the various classes of students were when the study began and the equivalence of the skills of the teachers. This means that your expectation that the researchers made sure that the groups were equivalent and that the teachers were comparably expert before the study began was unfulfilled. Proceeding, you anticipate a valid and reliable measure of the dependent variable, reading ability. But instead of finding that standardized reading tests were used, an administrator makes rankings of the various students, and testimonials are given about the accuracy of this ranking procedure. Turning to the analysis of the data, you expect, in view of the *ordinal level of measurement*, that *nonparametric statistics* will be used. Again, your expectancies are not met; the authors use *parametric statistics* instead. Curious about how all these issues are handled in the Discussion section, you find unjustified authoritative pronouncements and unfounded causal statements in support of the hypotheses.

How different is this critical reading process from the passive-receptive mode? In the passive-receptive mode, you might see a scientific research report as excessively detailed and would much prefer to read the abstract and skip the rest. Good “critical” interactive reading does not *require* the uncovering of crucial flaws. A special pleasure and admiration come to you when expectations are met. Indeed, there is little that is more satisfying to the critical reader of science than when expectations are exceeded or when the researchers anticipate and forestall the problems that you have thought about (or even had not considered), when the research procedures not only fit basic requirements (meet the standard on the yardstick) but do so in a particularly clear and original way, where the data are analyzed in an especially convincing manner, where the inferences from the data are unquestionably sound, and where the discussion is impressively profound and thought provoking.

When you critically read sound scientific material, you will say, “I’ve learned something and will incorporate it into my knowledge base; I’ll accept the conclusions and will cite them and apply them when appropriate.” When you critically read a flawed study, you cannot accept any of it as true. Exhibit 1.1 summarizes the key characteristics of critical reading.

KNOWLEDGE OF RESEARCH DESIGN

You may sit down in a state of heightened alertness, eager to critique an article of interest. Are you prepared? Do you know what to look for and what to focus your

EXHIBIT 1.1**Key Characteristics of Critical Reading**

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- Actively anticipate what you expect to read.
 - Use explicit rules to judge whether your expectations are met.
 - While reading, constantly monitor whether your expectations are being met, not just when you are done reading.
 - Have clear and justifiable reasons when you make an evaluative judgement about the strengths and limitations of an article.
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attention on? Will you see the defects or be unable to detect them? If you do not have the training to critique the article intelligently, you are bound to have concerns before you begin. You are in a position analogous to that of a person who is contemplating the purchase of a used car. Should faith be placed in the salesperson's appraisal of the car's merits? Do you know how to assess it? Do you know all the potential problems that may be present in the car's mechanical and electrical systems and how to evaluate their presence or absence?

People who lack specialized knowledge can always hire an expert to advise them. For example, you may rely on drama and motion picture reviewers to offer suggestions about what is worth seeing, or you can use websites that accumulate film reviews by experts and untrained movie-goers, a more democratic method of knowing.

PEER REVIEW

When it comes to scientific journal articles, there are usually no reviews to consult, though research that generates attention may have comments written on it that appear with the article later in the same journal or on the blogs of other researchers. Also, articles in some journals have been prescreened by referees and an editor. This is called *peer review*. Peer review can be described as "a formal system whereby a piece of academic work is scrutinized by people who were not involved in its creation but are considered knowledgeable about the subject" (Wager, Godlee, & Jefferson, 2002, p. 3). For master's theses and doctoral dissertations, the committee overseeing the awarding of the degree serves as the peer reviewers.

In a journal that uses peer review, the editor selects reviewers (often two, three, or four) because of their expertise in the substance and methods of the manuscript being evaluated. It is meant to ensure the quality of reports that appear in the scientific record. Peer review is used to evaluate the quality of reports submitted for journal publication, proposals to present at professional meetings, grant applications, reports, and book proposals and manuscripts.

You might think of peer review as the inspection a car dealer says was conducted on a used car before it is offered for sale. Although we do not want to compare science editors with used car dealers, it is important to make the point that such inspections may have occurred but should not lead you to complacency. You need to inspect and test-drive the car yourself.

Also, there are many journals that do not perform peer reviews before they publish articles. The number of these journals is proliferating on the Internet. Be careful here. Your first obligation as a critical reader is to discover whether the journal conducts peer review. Journals also can have more or less stringent peer review criteria before they publish an article. Journals published by the American Psychological Association (APA) have among the strictest requirements for publication. In 2015, the rejection rate of articles submitted for publication in APA (2016) journals averaged 70% and ranged from 45% to 90%. Of course, this rejection rate is based only partly on the quality of the research. It also includes article submissions that were deemed inappropriate for the journal, were not well-written, or do not make enough of a contribution for publication in a prestigious journal. Any rejected article can be revised and submitted to a different journal.

Regardless of how stringent this appears, no article that succeeds in being published is accompanied by a guarantee of excellence. No research is perfect, nor does any research meet all the standards on the yardstick. Flaws and weaknesses appear in all published articles. Critical reading allows you to decide where these flaws are, how serious they are, and whether they lead you to accept or reject the assertions contained in the article. And the existence of flaws should not discourage you regarding the value of conducting and critiquing research. The best-established scientific conclusions occur when evidence accumulates, when multiple studies, none perfect, lead to the same answer.

Any research article has three components: (a) the context and purpose of the research as reflected in the literature review, statement of the problem, and formulation of hypotheses; (b) the research itself; and (c) the interpretation and discussion of the results. A reader who knows much about a topic, but little about research design principles, is not in a good position to evaluate a research article critically. A reader who knows little about the topic, but much about research design, can nonetheless evaluate the design and execution of the research. Such a reader may miss some potentially crucial details regarding the choice and use of specialized apparatus or the selection of the best available measures or most advanced techniques. Generally, however, readers who have knowledge of methods can review material that is within their general sphere of knowledge.

The principles of research design transcend content areas. The effectiveness of a reader is therefore dependent on knowledge of research design and on skillful application of that knowledge. The remaining chapters in Part I discuss those aspects of research design that are essential to know if you are to conduct a comprehensive appraisal of a research article. A systematic approach to the critique of research reports is outlined in the introduction to Part II.

Research Questions and Hypotheses

2

A critical reader of research

- knows the different types of research questions;
- understands that different research questions require different types of research designs to answer them—if the question and design do not correspond, the conclusions of the research are suspect;
- requires that the evidence meet stringent conditions to assert a causal relationship between variables—these conditions are never fully met, just more and less so, and experiments are the best way to test for cause–effect relationships; and
- looks for hypotheses that are (a) clearly and accurately linked to the research problem and its historical context, and (b) stated in operational terms that make clear statements about what the researchers think the is answer to the research question.

In this chapter, we examine the principal types of research questions or problems, give examples of each, and review the design expectations that each type of question establishes. Then we address the complex matter of causation and raise issues that must be resolved whenever the research question involves a causal relationship. Finally, we discuss the importance of defining the terms involved in the research and formulating clear hypotheses.

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